Military aviation

By the early 21st century, air power was seen as the key to power projection and security, suiting citizens in technologically advanced democracies accustomed to peace and reluctant to countenance the level of casualties that a ground war usually involves. The 2003 invasion of Iraq by American and British forces and the subsequent occupation of the country, as well as the worldwide threat of terrorist attacks, became the focus for thinking about future war plans, although planners ran the well-known risk of preparing for the last war rather than the next. Air operations in the 2003 Operation Iraqi Freedom showed how technologies had been refined, extended, and advanced to a far higher pitch of efficiency since the 1991 Gulf War.

"Risk-free" air attack

Precision guided bombs and missiles had constituted less than 10 percent of munitions used during the Gulf War; in 2003, around two-thirds of airborne munitions were "smart." The use of JDAMs (Joint Direct Attack Munitions) was crucial. A JDAM was a bomb with a guidance package linked to a Global Positioning System (GPS) or Inertial Navigation System (INS). The map coordinates of a target on the ground were fed into the bomb's control system. Once it was released from the aircraft, the JDAM would then steer itself to the designated coordinates without any further guidance. JDAMs were found to be accurate to within around 33ft (10m) and could be used in bad weather conditions that would create problems for laser- or TV-guided munitions.

Other key factors in the 2003 invasion included the increasing use of UAVs (Unmanned Aerial Vehicles), more than 100 of which were deployed. They mostly carried out reconnaissance missions, but Predators on occasion acted as UCAVs (Unmanned Combat Air Vehicles), executing precision air strikes with Hellfire missiles. The introduction of "time-sensitive targeting" – the direction of strike aircraft lurking over the battle area on to a target only minutes after it was identified – was considered a success. In one incident, a B-1 bomber struck a house where Saddam Hussein and his sons were said to be meeting within 12 minutes of the target being identified by US intelligence.

Although a number of well-known problems resurfaced during the Iraq invasion, including civilian casualties caused by occasional inaccuracy or mistaken target identification and losses to "friendly fire," the effectiveness of the deployment of air power could not be disputed. Iraqi ground forces simply could not operate without air cover. Command and communications could not be maintained in the face of precision air strikes. With its massive costs and its immense organizational and technological demands, air power remained the clearest determinant of power differences between states in the early 21st century. Iraq might have given a superpower a fight on the ground, but in the air, it could not compete.

Limited usefulness

The subsequent occupation of Iraq graphically illustrated the limitations of air power. Operating against an enemy indistinguishable from the rest of the local population under circumstances in

IRAQ CAMPAIGN MEDAL
Even high-tech campaigns such as Operation Iraqi Freedom involve risk to ground troops and air crews. Among the recipients of this medal were air crew who had flown sorties over Iraq on 30 consecutive days.

SHOCK AND AWE
During the Iraq invasion, media attention was riveted by the spectacle of strikes by aircraft and cruise missiles on Baghdad. Although precisely targeted against official buildings, these air strikes were linked to the fashionable military principle of "shock and awe" – the use of a spectacular display of power to undermine the enemy's will to fight.